

Evaluating the Effect of Chunking Strategies on Retrieval and Generation Performance in RAG Systems

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Introduction

Retrieval Augmented Generation (RAG) has become an increasingly prevalent technique in the space of large language models, allowing for precise context augmentation from within large text corpora without exhausting context length. In place of appending whole documents to the prompt, a subset of relevant chunks is retrieved through the comparison of embeddings, which are then provided as context to the model accompanying the user query, allowing robust scaling while retaining comprehension of fine details. Nonetheless, the effectiveness of this procedure hinges on the quality of the chunks that are supplemented, a factor influenced by numerous variables. In particular, the method by which documents are split into chunks, which alters the size and boundaries of the information accessible to the model, impacts both retrieval accuracy and the ability of the model to utilize the context. With the significance of chunking established, an important question arises: which chunking strategy is most conducive to accurate RAG performance? While retrieval metrics are a common tool in evaluating chunking methods, their results may not necessarily correlate with the final generated answer quality. We examine three common chunking techniques and assess their effect on retrieval precision as well as the quality of generated answers, calling attention to a discrepancy between the two targets.

Methods

The typical baseline for chunking experiments is the fixed-length approach, in which documents are split into chunks of a fixed token size, and include a fixed token overlap in order to minimize loss of context between consecutive chunks. 500 tokens with an overlap of 50 tokens serves as a moderate setting, reasonably representative of the method as a whole. The foreseeable issue presented by fixed-length chunking is that chunk boundaries may lie between sentences, compromising the continuity and, therefore, accessibility of information. Although the aforementioned overlap is a partial remedy, the issue persists so long as chunk boundaries are unregulated. Sentence-based chunking is an attempt to thoroughly mitigate the defect by splitting documents exclusively at sentence boundaries, with each chunk containing a fixed number of sentences, as well as a single-sentence overlap. A consistent chunk size of 5 sentences was chosen for this method, encapsulating a similar volume of information per chunk as with the previous method, albeit with greater variation. However, the possibility of greater improvement upon sentence-based chunking still remains, as the method fails to consider implicit boundaries of text transcending the form of sentences. As such, sentence-based chunking may split sections of text discussing a single topic, fragmenting its availability during retrieval and generation. Semantic chunking presents a resolution by utilizing embeddings to identify shifts in topic throughout a body of text. Each contiguous subsequence of three sentences is compared to the following subsequence, utilizing their respective embeddings in order to produce a numerical similarity (BAAI/bge-base-en-v1.5 was used for this intra-chunk comparison, the same model used for inter-chunk comparison during retrieval). The corresponding points in the text at which the similarity drops below a threshold are the

points at which chunk boundaries lie. Note that by the design, each pair of contiguous subsequences will contain two common sentences. This ensures that minute shifts in tone or structure are disregarded in favor of the broader shifts in topic that the procedure is intended to capture. As for the similarity threshold, it was determined by collecting similarity values across the entire test corpus. The 20th percentile value, 0.85, was selected to roughly emulate the previous chunks of 5 sentences. All three chunking methods extend previous variations, allowing decisive tradeoffs to be revealed through experimentation.

The dataset on which the chunking strategies were evaluated consists of 10 papers from the Social Sciences journal, spanning a variety of topics. For the purpose of assessing the robustness of textual chunks, social science presents itself as an appropriate field, as papers are predominantly made up of text. Any tables or graphs present in the original papers were safely omitted, as their inclusion played no compulsory role in communicating the intended messages. All ten papers were chunked by each of the three methods, with the resulting chunks of each method being stored in an isolated vector store. Consequently, at retrieval time, the vector store of each method contained chunks irrelevant to the prompt as well as the desired relevant chunks, effectively allowing evaluation of the ability of relevant chunks to delineate themselves in the midst of irrelevant chunks.

The prompts used in evaluation were produced directly from the papers, with one from each of the following question categories for each paper, yielding five questions per paper, fifty questions total. The standardization in prompt rigor rewarded accuracy and flexibility in retrieval.

- Fact-based: requires reciting information that is explicitly stated in the text
- Methodology: requires identifying elements of the methodology of the paper
- Results: requires summarizing the results demonstrated in the paper
- Synthesis: requires synthesizing multiple sections of text in order to generate a conclusion
- Analytical: requires consolidating the main ideas of the paper in order to draw abstract conclusions

Furthermore, the answer to each question and the section of text in which it was located were included in the dataset. These ground truths were used in both phases of evaluation in order to preserve the integrity and applicability of later findings.

The chunks produced by each method were embedded using BAAI/bge-base-en-v1.5, marking the end of chunk production. The retrieval procedure began as follows, remaining unchanged for each method. The prompt was embedded and compared to the embeddings of each of the chunks in the vector store of the current method using cosine similarity. The chunks corresponding to the 5 most similar embeddings were selected as the most relevant chunks to the prompt. At this point, a consistent criterion was required to judge whether a retrieved chunk is equivalent to a ground truth chunk. This functionality is critical to the impartial evaluation of the aforementioned retrieved chunks. Chunks cannot be compared literally as strings because chunk boundaries differ between chunking methods, meaning that chunks encompassing identical information would still be judged as dissimilar purely as a result of different boundaries. Instead, chunks were reinterpreted as sets of bigrams in order to facilitate comparison. If the chunks shared more than two unique bigrams, and the shared bigrams comprised more than 30% of the ground truth chunk, then the chunks were deemed analogous in the information they carry, and the

retrieved chunk was considered relevant. This process of comparison underpins the functionality of each of the following metrics used in evaluating retrieval accuracy.

The use of numerous metrics yielded a holistic view of advantages and drawbacks, while avoiding the risk of allowing noisy outcomes to influence conclusive judgments based on a single metric. In turn, three distinct metrics were employed, each surveying particular aspects of retrieval accuracy. In the process reported above, chunks were retrieved in a ranked order, with the chunks corresponding to the highest cosine similarity falling closer to the beginning. Thus, the rank at which relevant chunks were placed serves as a robust measure of retrieval accuracy for instances where all methods yield similar chunks at retrieval. For each set of retrieved chunks, the reciprocal of the highest rank at which a relevant chunk was present was recorded rather than the plain rank in order to construct a zero-to-one scale, in which higher values correspond to greater results, ultimately creating the reciprocal rank metric. The following metrics, Recall@k and Precision@k, both operate primarily using the number of relevant chunks in the top k retrieved chunks, with k taking on a value of 1 and 5 for two separate evaluations. However, when computing Recall@k, this value is divided by the total number of relevant chunks identified in the ground truth, whereas in Precision@k, the value is divided by k, the number of chunks being evaluated. In turn, the two metrics are distinct representations of retrieval success, with Recall@k rewarding inclusiveness and comprehensiveness, and Precision@k incentivizing selective retrieval with fewer irrelevant chunks. This allowed a greater capacity for accurate conclusions when paired with the reciprocal rank metric.

In order to explore the significance of chunking strategies beyond retrieval, an evaluation of generated responses was conducted in addition to retrieval evaluation. With sets of relevant chunks preestablished for each prompt and each method, a supplementary prompt is the final primary component required before inference may be carried out by the chosen large language model, Llama-3.1-8B-Instruct. Such a prompt must incorporate the retrieved context, the prompt to be answered, and clear instructions as a means of enforcing appropriate model behavior and reducing confounds during generation. The model was directed to exclusively utilize the given context when responding to the prompt, operating without any external knowledge. This constraint is critical in order to ensure that the effectiveness of the language model does not mask the ineffectiveness of chunking. Furthermore, a consistent response was programmed for instances where the answer cannot be synthesized from the given context, helping bolster interpretability and clarity in the evaluation phase.

After obtaining generated responses for each of the prompts sourced from each of the chunking methods, an objective evaluation of the responses was carried out utilizing a second large language model in order to capture variation in the completeness and relevance of answers beyond the capabilities of simpler methods. However, attempting to produce numerical scores for each answer leads to noisy valuations lacking confident interpretability. The superior technique is ranking, in which the model is supported by specific criteria to arrange the given answers, with the validity of a particular answer existing only in relation to the others, producing greater reliability for conclusive judgments. In particular, gpt-oss-120b was instructed to prioritize factual correctness, followed by completeness and clarity when ranking answers in order to provide a realistic emulation of human judgement criteria. Note that a temperature of 0 was utilized as it is essential to defend against noise and inconsistency in the language model for evaluation tasks.

Results

The findings produced by utilizing both retrieval evaluation and generated-response evaluation demonstrate a comprehensible divergence in the factors that bolster retrieval quality in comparison to the factors that support robust generation. However, despite this divergence, a decisive judgement exists regarding the ideal method in the form of a balanced overlap between a multitude of characteristics: consistency in the size of chunks, adherence to topical shifts in chunk boundaries, and model interpretability.

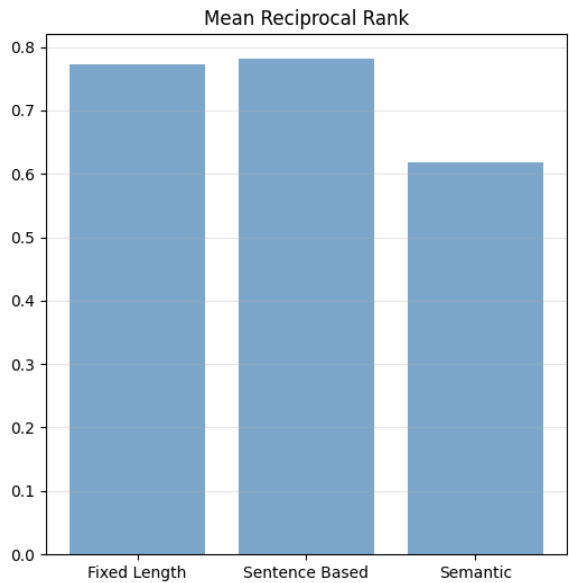


Figure 1: Mean reciprocal rank across chunking methods

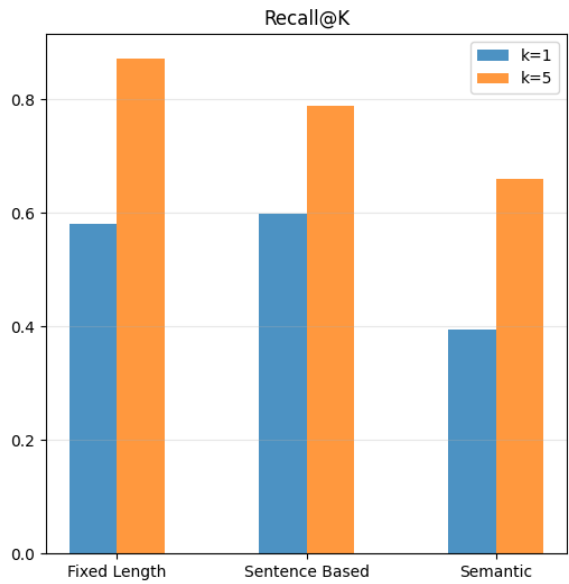


Figure 2: Recall@k across chunking methods

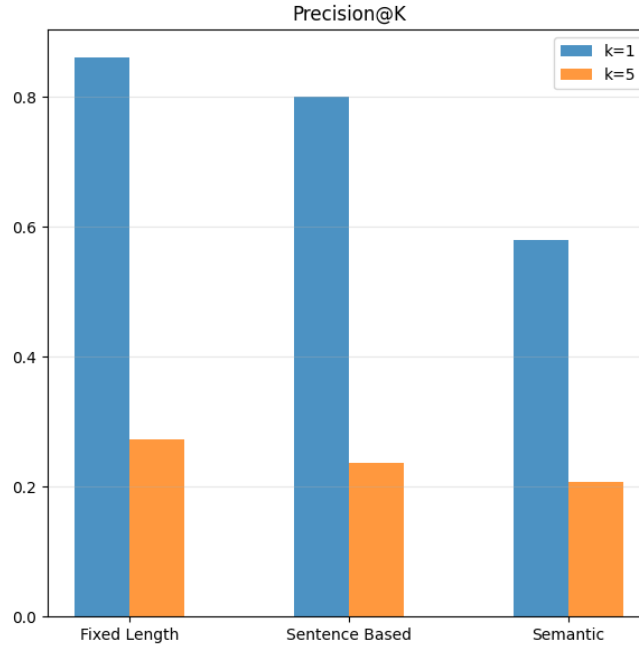


Figure 3: Precision@k across chunking methods

In inspecting the results of the reciprocal rank evaluation, semantic chunking clearly lags behind its simpler counterparts, displaying an average reciprocal rank of 0.62 in comparison to 0.77 and 0.78 for fixed-length and sentence-based, respectively (Figure 1). This pattern continues to manifest in the following tests of retrieval, with fixed-length chunking outperforming sentence-based chunking by a significant margin as well. For instance, in Recall@5, fixed-length exhibits an average score of 0.86, followed by 0.79 from sentence-based, and 0.39 from semantic (Figure 2). These results replicate themselves in the same manner for Precision@k, suggesting that the intelligent chunk boundaries offered by semantic chunking undermine a deeper fundamental necessity for effective retrieval: chunk size consistency (Figure 3). By implementing chunk boundaries that shift with the subject of the text, the size of chunks no longer depends on an external rule, as it does with the other two methods, leading to larger variation in size between chunks. In contrast, fixed-length and sentence-based strategies produce chunks of roughly equal size by design, with fixed-length being slightly more consistent than sentence-based, as it even ignores sentence boundaries, which vary in size, albeit marginally. This explains the gap in retrieval performance between the three methods and establishes a clear principle for effectiveness for retrieval metrics: consistency in chunk size.

Nonetheless, the quality of generated answers still remains the principal criterion in determining the optimal chunking method. While the trends revealed in this component of evaluation do not completely overturn the supposed conclusions drawn from retrieval evaluation, they most certainly defy the notions in a clear manner, adding nuance to the central decision.

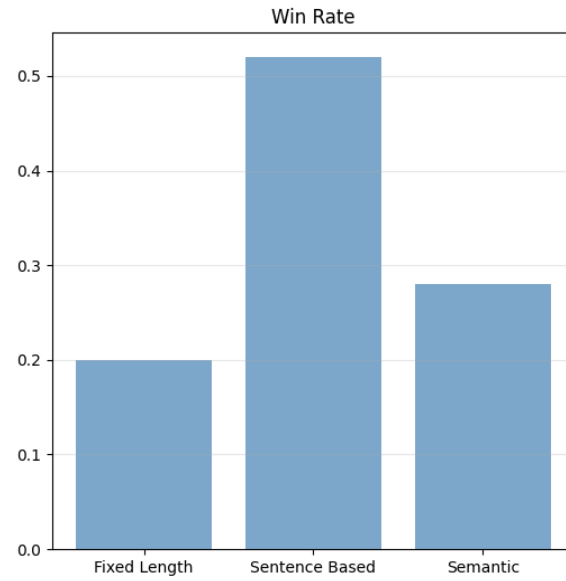


Figure 4: Win rate of each chunking method

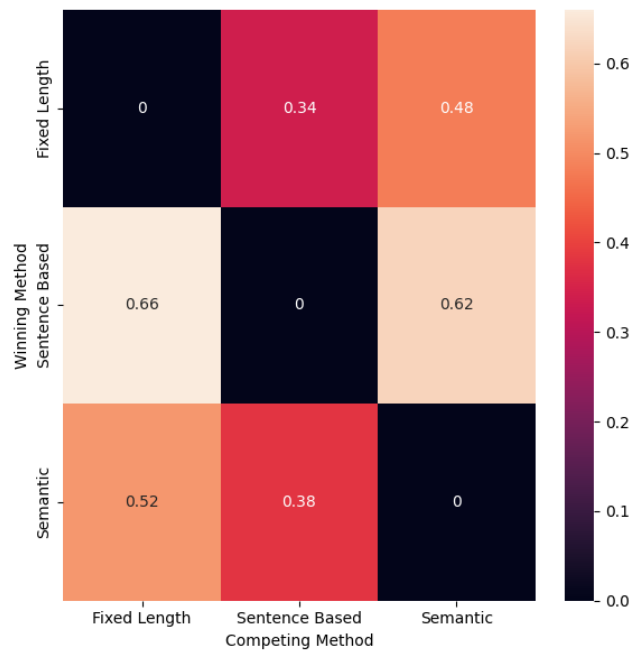


Figure 5: Pairwise comparison matrix

Fixed-length chunking, which dominated the retrieval metrics, lags behind the other two methods when generation is considered, achieving the best answer 20% percent of the time. In contrast, semantic chunking, the least effective during previous tests, yielded the best answer 28% of the time. The method that realized the most robust responses consistently was sentence-based chunking, doing so 52% of the time (Figure 4). The stark dissimilarity manifests itself due to a shift in the characteristics being evaluated, and reveals the superior method among the three being evaluated. The former portion of the evaluation, albeit functional, was inherently biased towards strategies prioritizing coverage over structure.

Yet semantic chunking, the method that is said to align most closely with semantic structure, still did not emerge as the leading approach in the latter portion of evaluation, suggesting that a balance of coverage and structure is required in order to facilitate the generation of accurate and complete responses. Sentence-based chunking appears to replicate the consistency in chunk size offered by fixed-length chunking, while maintaining coherence at chunk boundaries by utilizing sentences as endpoints, allowing it to surpass the other methods.

Conclusions

As demonstrated by both segments of testing, retrieval evaluation and generated-response evaluation, the result varies significantly as various factors are taken into consideration in evaluation. Although retrieval evaluation appears to assess the extent to which answers are present in the retrieved chunks, originally thought to be the prime limiter in RAG performance, later tests reveal that the success of the pipeline is multifaceted, depending on the chunking strategy in more than one way. Chunks must not only contain the answer to the given prompt, but they must represent the information in a manner that is easily accessible to large language models, as exemplified in the success of sentence-based chunking over fixed-length in generative tests. All in all, these findings indicate the incomprehensiveness of retrieval metrics alone as proxies for downstream success in the RAG sequence, underlining the importance of end-to-end evaluation in order to yield results truly illustrative of performance.